

On-Premise on a Raspberry Pi 5

One-time hardware purchase, kept on your own premises, published to the internet through your own domain and a free Cloudflare Tunnel. Sized for a single institute of 200 to 300 users.

ONE-TIME SETUP

₹26,000

approx. US\$ 275

RUNNING COST

₹350 / month

approx. US\$ 3.70 per month

TOTAL, 5 YEARS

₹47,000

approx. US\$ 495

1. One-time investment

Item	INR	USD	Why it is needed
Raspberry Pi 5, 8 GB	7,500	79	The server itself
Official 27 W power supply	1,700	18	Stable power
Active cooler and case	1,600	17	Runs 24 × 7 without throttling
NVMe adapter and 500 GB SSD	5,300	56	Fast, reliable storage (not an SD card)
Spare boot microSD card	700	7	Recovery media
Small UPS, 600 VA	3,700	39	Survives short power cuts
External 1 TB backup drive	4,500	47	Local nightly backup
Setup and installation labour	1,000	11	One-time, if outsourced
Total one-time	25,900	274	

Prices are indicative Indian retail as of August 2026 and converted at ₹95 per US\$.

2. Recurring cost

Item	Per year (INR)	Per year (USD)
Domain name	1,200	13
Cloudflare Tunnel and DNS	0	0
Electricity (about 6 units a month)	600	6
Off-site backup storage, 50 GB	1,800	19
Internet connection	0	0
Total per year	3,600	38

Assumes the institute's existing broadband line is used, so there is no additional connectivity charge.

3. Cost over time

Period	INR	USD
Year 1 (hardware + running)	29,500	310
Year 2	3,600	38
Year 3	3,600	38
Years 4 to 5	7,200	76
Hardware refresh allowance	3,000	32
Five-year total	46,900	494

For comparison, the same workload on a public cloud costs roughly ₹5,700 per month, which is about ₹3.4 lakh over five years. See the companion cloud comparison sheet.

4. What you gain

- No monthly cloud bill. Costs are known and fixed.
- All institutional data stays physically on your premises.
- No per-user or per-record charges as the institute grows.
- Cloudflare Tunnel publishes the site securely at no cost, with no static IP needed.
- Pays for itself against cloud within roughly five months.

5. What you take on

- If the building loses power or internet beyond the UPS window, the system is offline.
- One machine, so a hardware failure means downtime until it is replaced.
- Someone on staff must confirm that backups are running.
- Comfortable for a few hundred users. Beyond roughly 500 concurrent users, move to cloud or a larger server.

Recommended for this client. At 200 to 300 users the workload is small, the cost difference is large, and the trade-offs above are manageable. A second identical Pi kept as a cold spare adds about ₹9,000 and removes most of the hardware risk.